Will to Power: Technoeconomic comparison of DC- and AC-coupled solar microgrids, natural gas, and grid AI data center power

AI OPEN DATA CENTER



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Background and motivations

- AI load is growing and large; ~half of total energy, most of future growth (Smith et al. 2026)
- Data centers are well suited to direct current power systems (Baranko et al. 2024; Tschudi et al. 2008; Ollas et al. 2023; Sandelic et al. 2019; Lo Franco et al. 2021, Hirose et al. 2018, Chen et al. 2022)
- Firms consistently express strong speed preference, but we don't have quantitative values
- Buildout colliding with supply bottlenecks for turbines (Anderson 2025), transformers, and slow grid interconnection
- Firms adopting exclusive PPAs (Elliott 2024; Mandler 2024) and behind the meter generation (Anderson 2025)

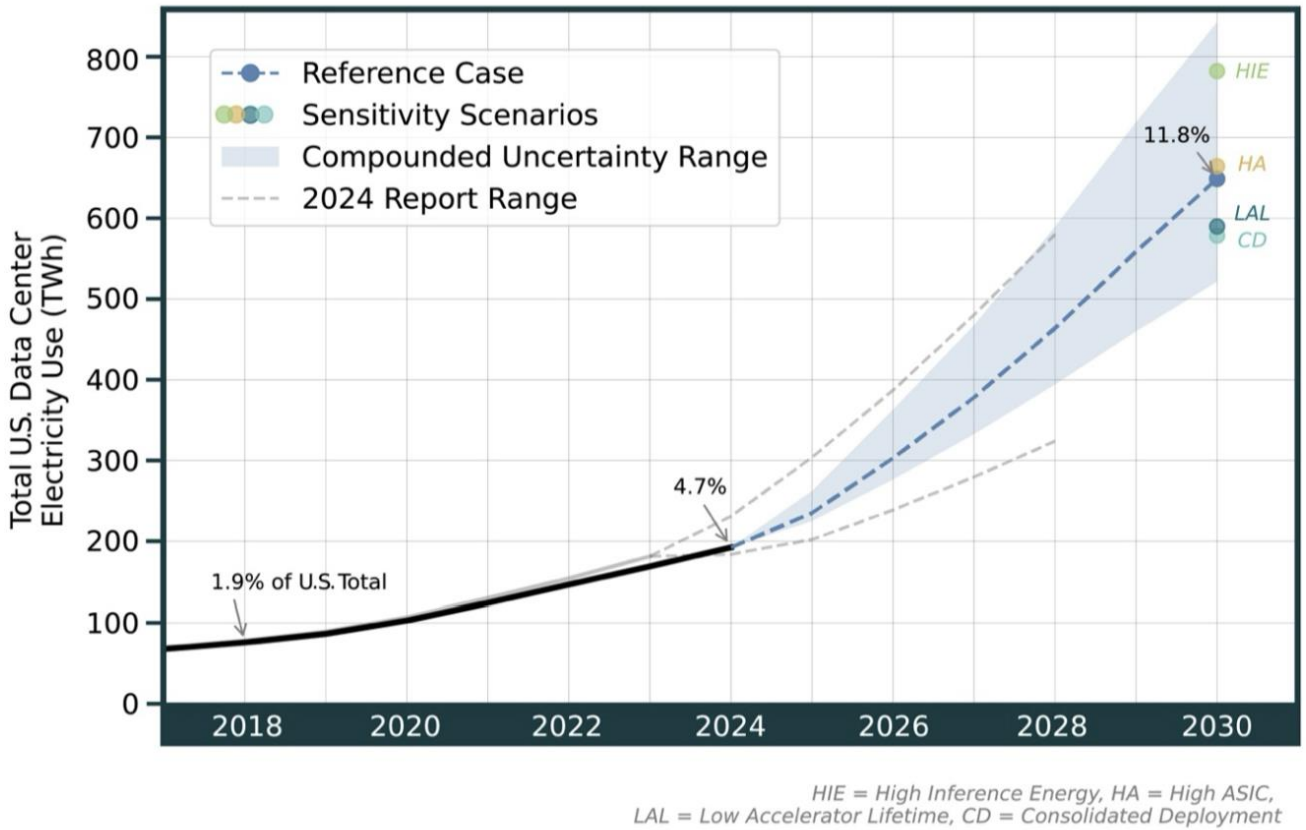


Figure 1. Total U.S. data center electricity use from 2017 through 2030

Smith, Sarah, Alexander Hubbard, Alex Newkirk, et al. 2026. *United States Data Center Energy Usage Report: 2025 Update*. Lawrence Berkeley National Laboratory. https://doi.org/10.71468/P1RP4F.

Research Question and Modelling Goals

Core Research Question

What is the technical feasibility and economic performance of islanded power systems to supply AI training datacenters across the United States?

Microgrid Considerations

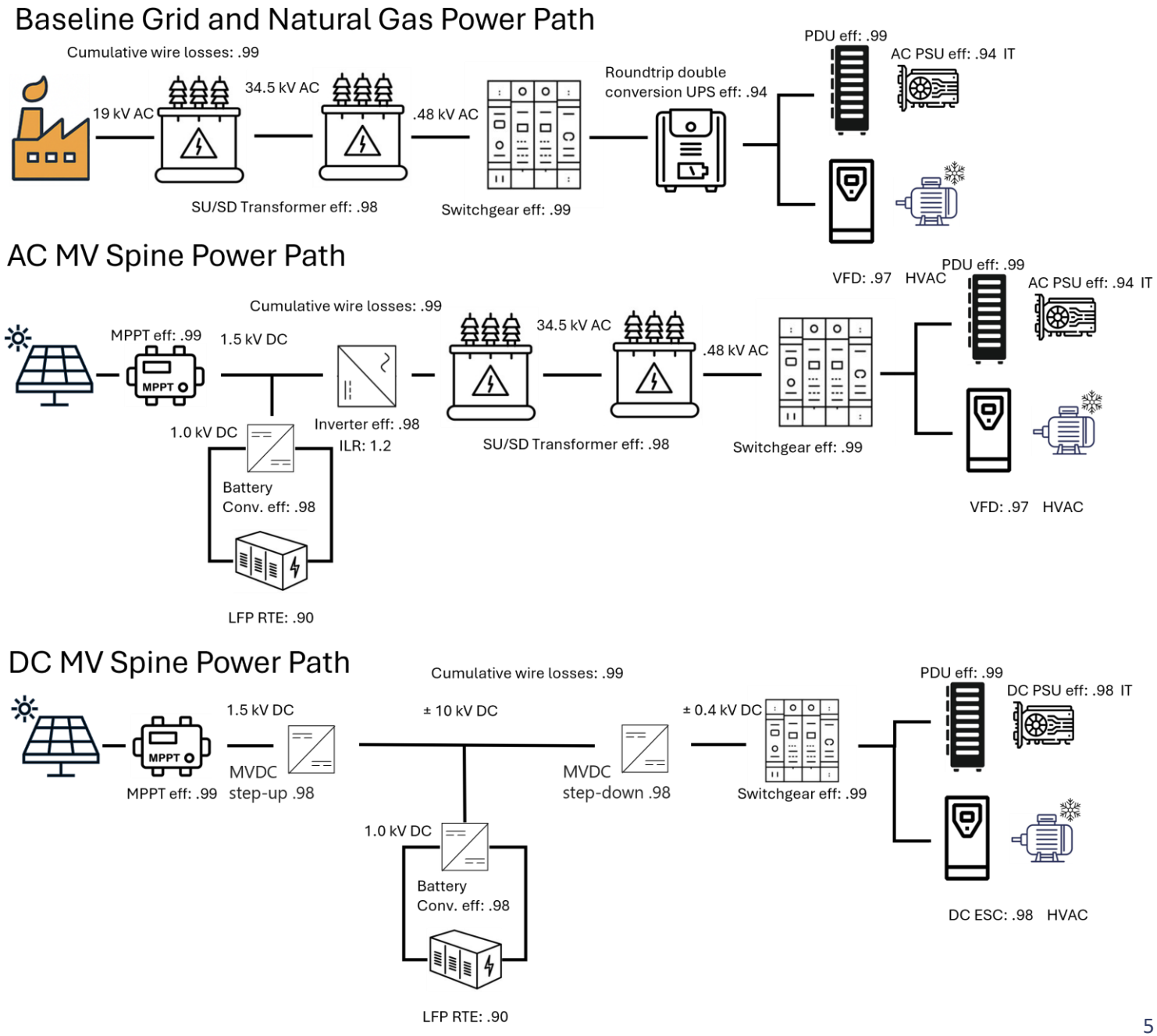
- Full-Stack: How would physically accurate microgrids (PV+BESS; NG) perform, across geographies and facility scales, under equivalent assumptions?
- Speed-to-Power: What is the marginal value of deployment speed?
- Efficiency: How do AC and DC power systems compare?

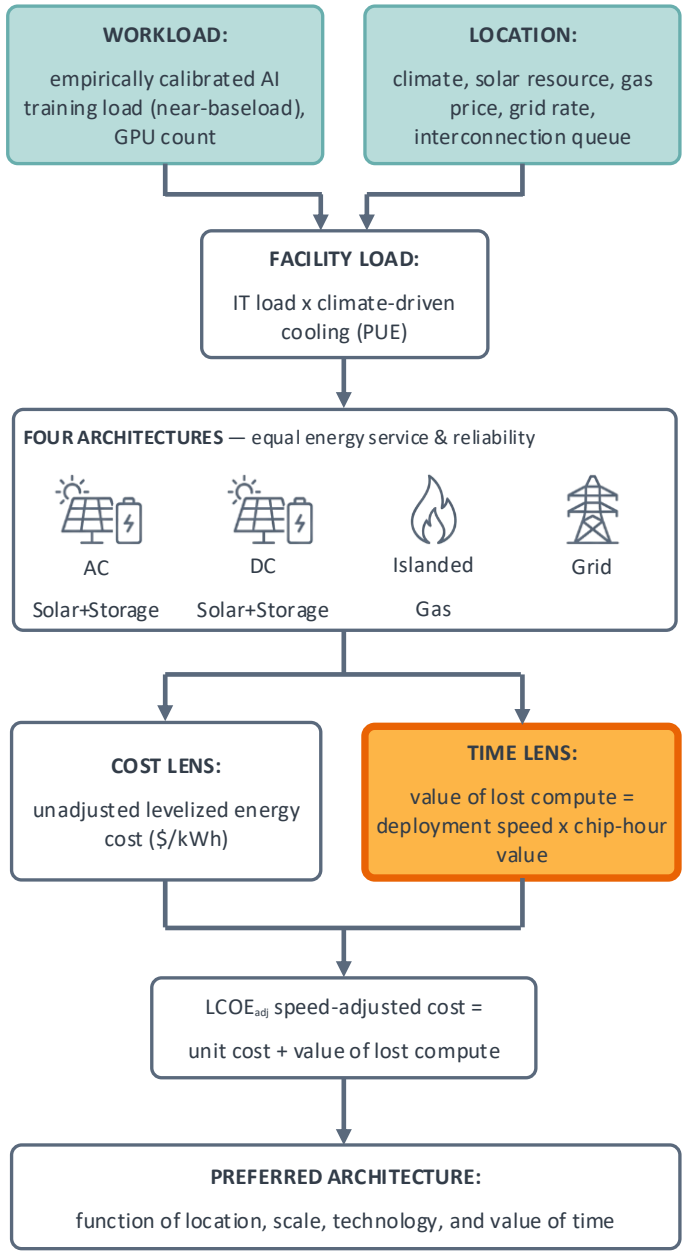
Methods and Data – Fair technoeconomic comparison incorporating deployment speed

Research Sub-question	Method	Data
How do AI training facility power needs vary by location?	Hourly facility demand simulation, Climate-dependent cooling modeling	Node-level IT demand model, Location-specific TMY data, cooling technology performance model (Lei and Masanet 2022)
How do microgrid architectures compare for meeting AI facility demand?	Technoeconomic modelling, mixed-integer optimization	Baranko et al. (2024) solar/storage tool, Smith et al. (2021) degradation model, NREL ATB 2024 system component costs, EIA regional energy prices
What efficiency gains do all-DC microgrids provide?	Power systems engineering, empirical efficiency benchmarking, topology performance comparison	Power electronics datasheets and benchtests (Gerber et al. 2023), high-resolution training workload time-series (Newkirk et al. 2025)
How does time-to-operation affect economic comparisons?	Time-value analysis of energized capacity, construction timeline modeling	LBNL interconnection queue data (Rand et al. 2024), expert interviews on microgrid construction, GPU spot market rental index price

Modeled Power System Architectures

- DC Microgrid designed to the ±400V OCP standard. Baseline modeled cases for all architectures employ medium voltage (MV) campus distribution spines stepped down at each data hall.





Tool Design & Speed-Premium

- Levelized-Cost-of-Energy (LCOE) at baseline reflect unit energy costs
  - TCO (sum of present value of costs) divided by discounted lifecycle energy (E_t)
- GPUs energized sooner generate value sooner
  - Slower to construct (or connect) power systems incur an opportunity cost over faster ones for each marginal idle hour (H_{idle})

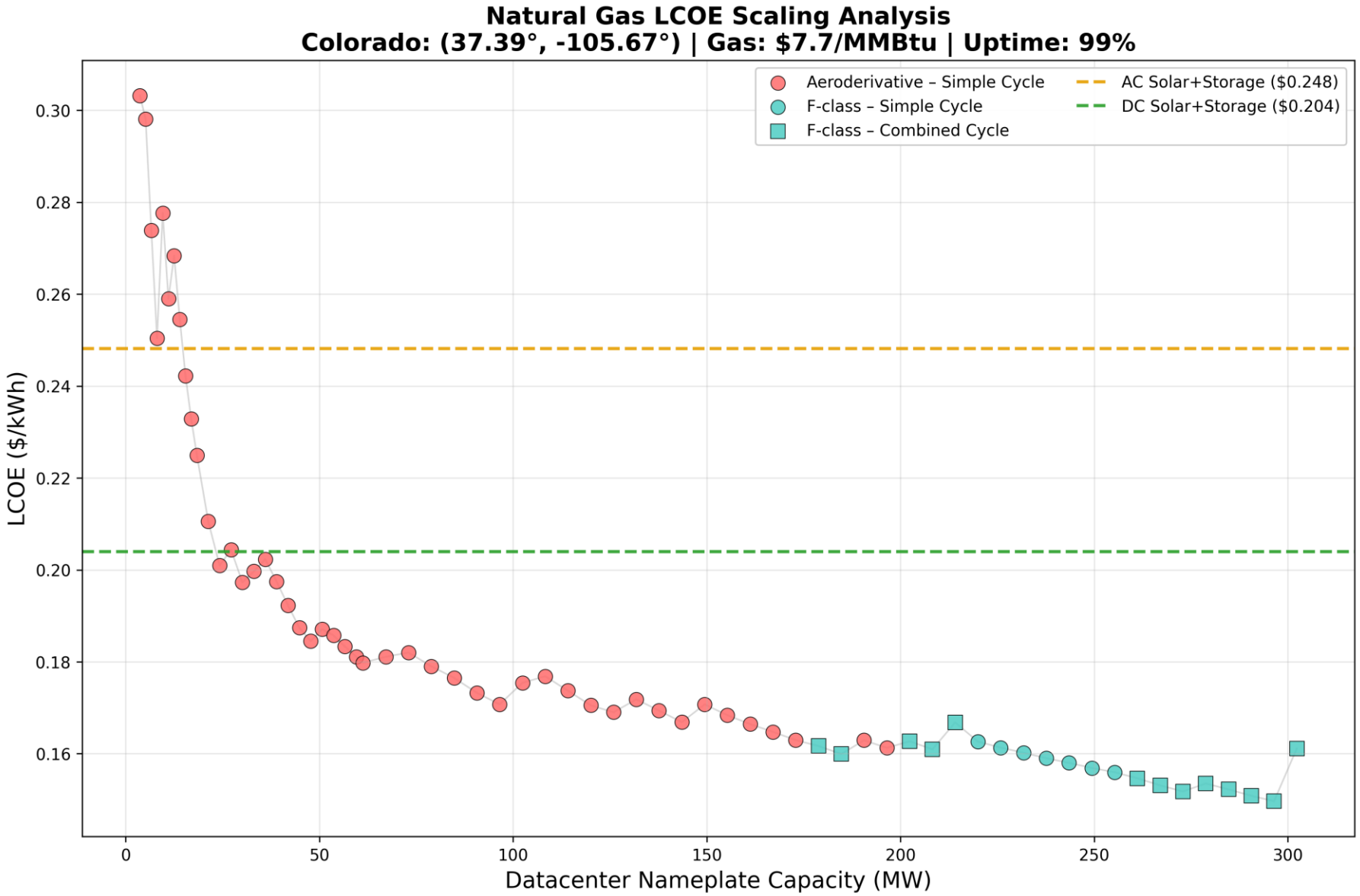
$$VLC_t = \sum_{h=1}^{H_{idle,t}} N_{GPU} \times p_{spot}$$

- The value of lost compute (VLC) is this foregone compute value
- GPU spot rental price index provides our baseline value (P_{spot}) for each individual GPU (N_{GPU})
  - This can be thought of as the floor-utility of an internal workload for a mixed model developer/infrastructure provider (e.g. Google)

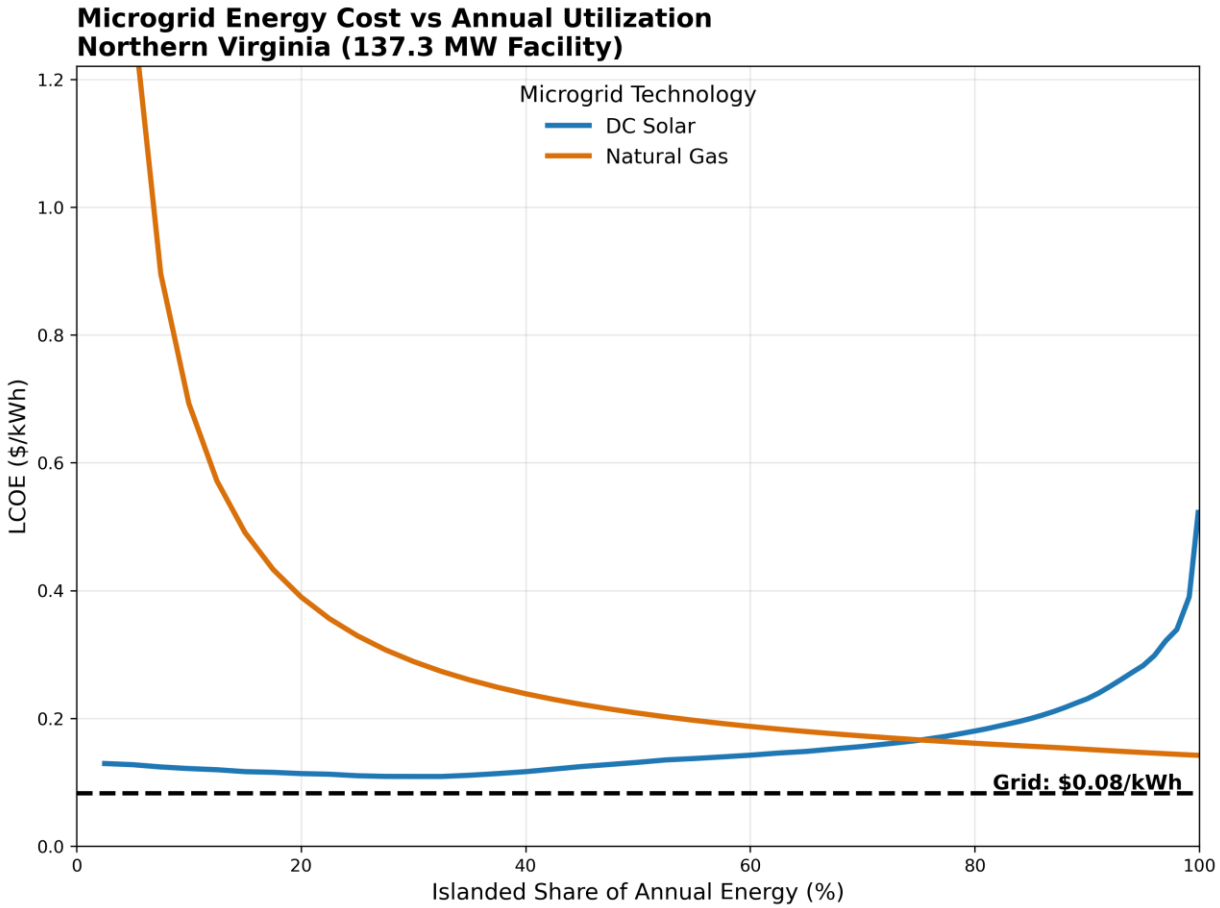
$$LCOE_{\text{adj}} = \frac{\sum_{t=0}^{t} \frac{\left(C_{\text{Cap},t} + C_{\text{Op},t} + VLC_t\right)}{(1+r)^{t+0.5}}}{\sum_{t=0}^{t} \frac{E_t}{(1+r)^{t+0.5}}}$$

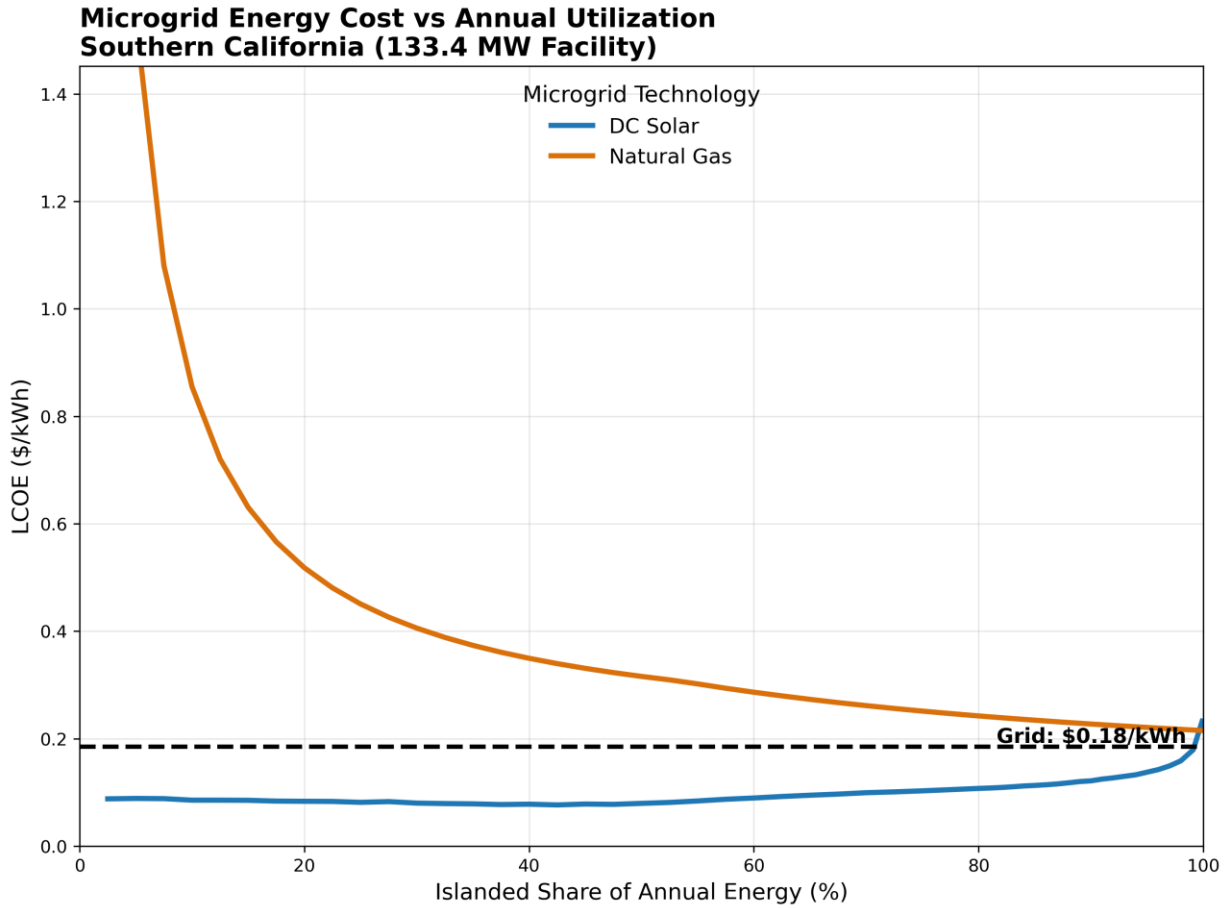
- $LCOE_{adj}$ integrates VLC alongside capital (C_{cap}) and operating costs (C_{op})

Results: natural gas costs are non-linear in the capacity

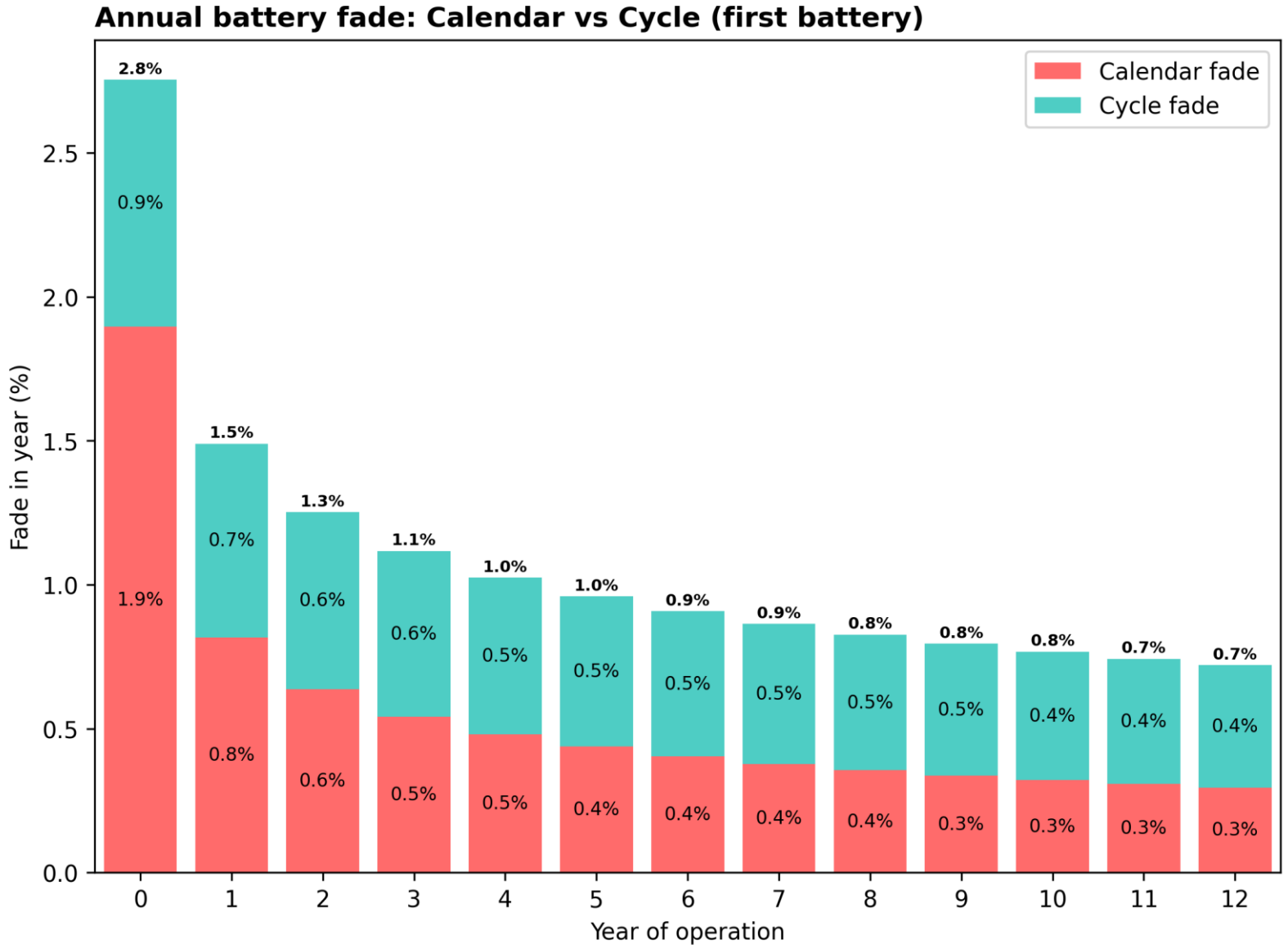


Results: “Last mile” of uptime in PV systems is expensive, but performance is location-specific

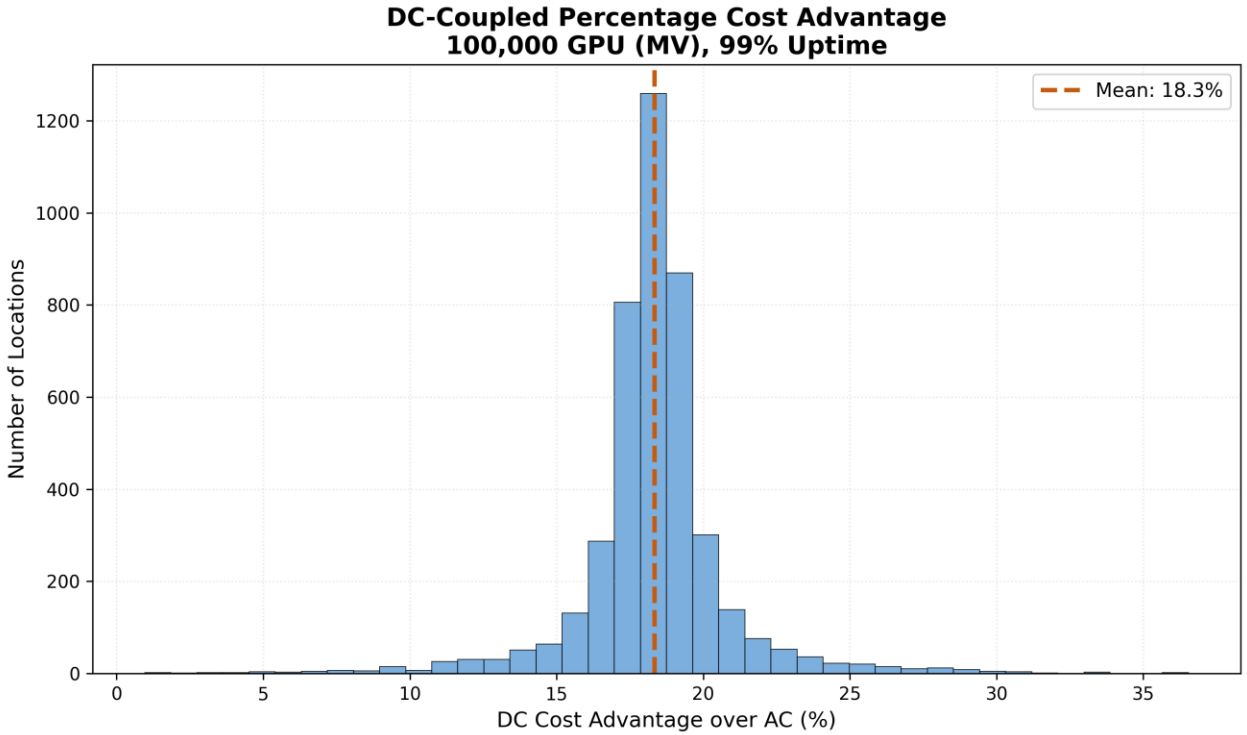


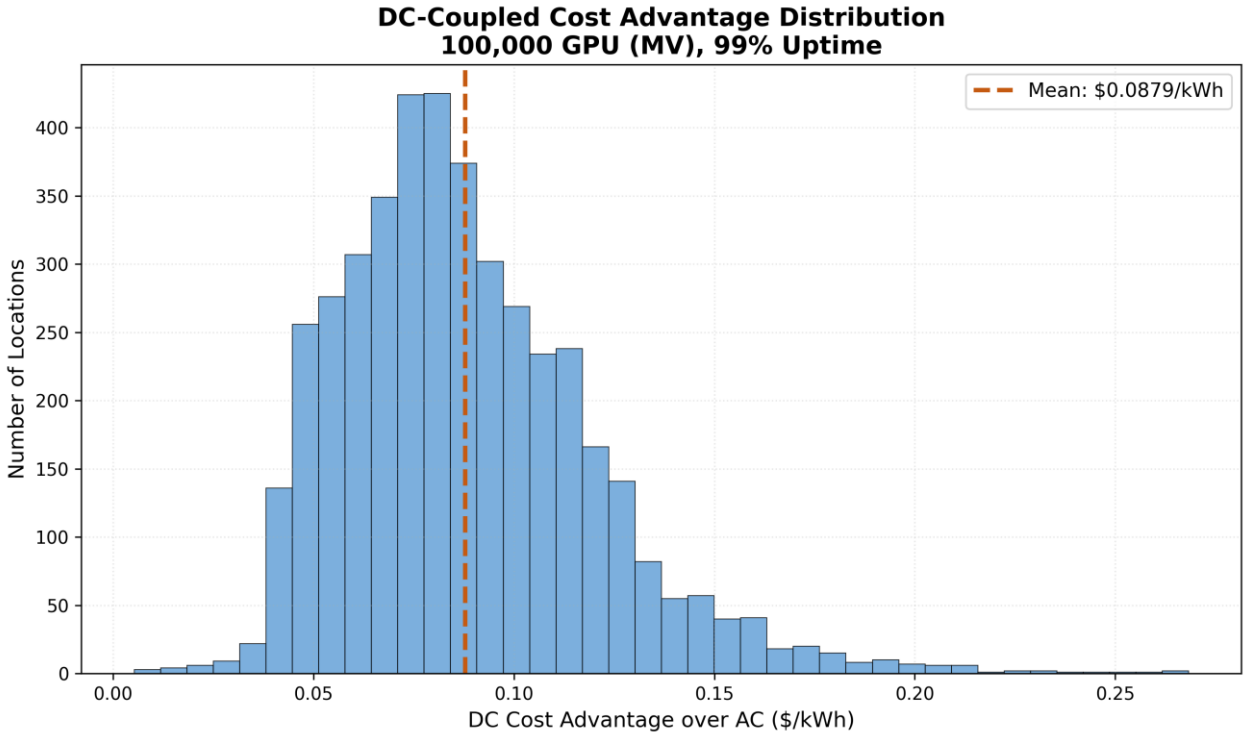


Results: duty cycle of modeled battery systems introduces in only modest capacity fade (system retains 89% initial capacity at Y13 replacement on average)



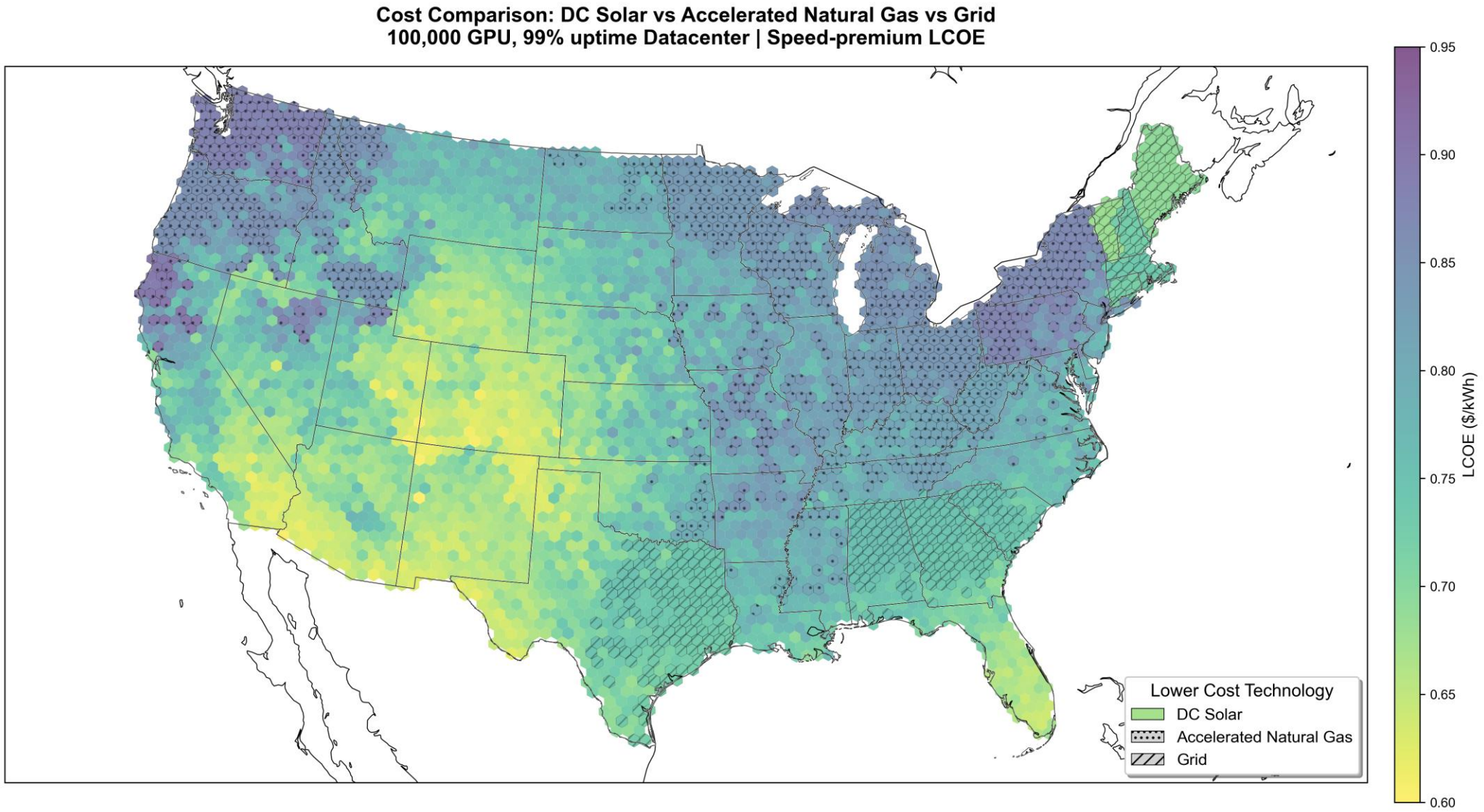
Results: all-DC performance improvement – 18% reduction in energy cost on average



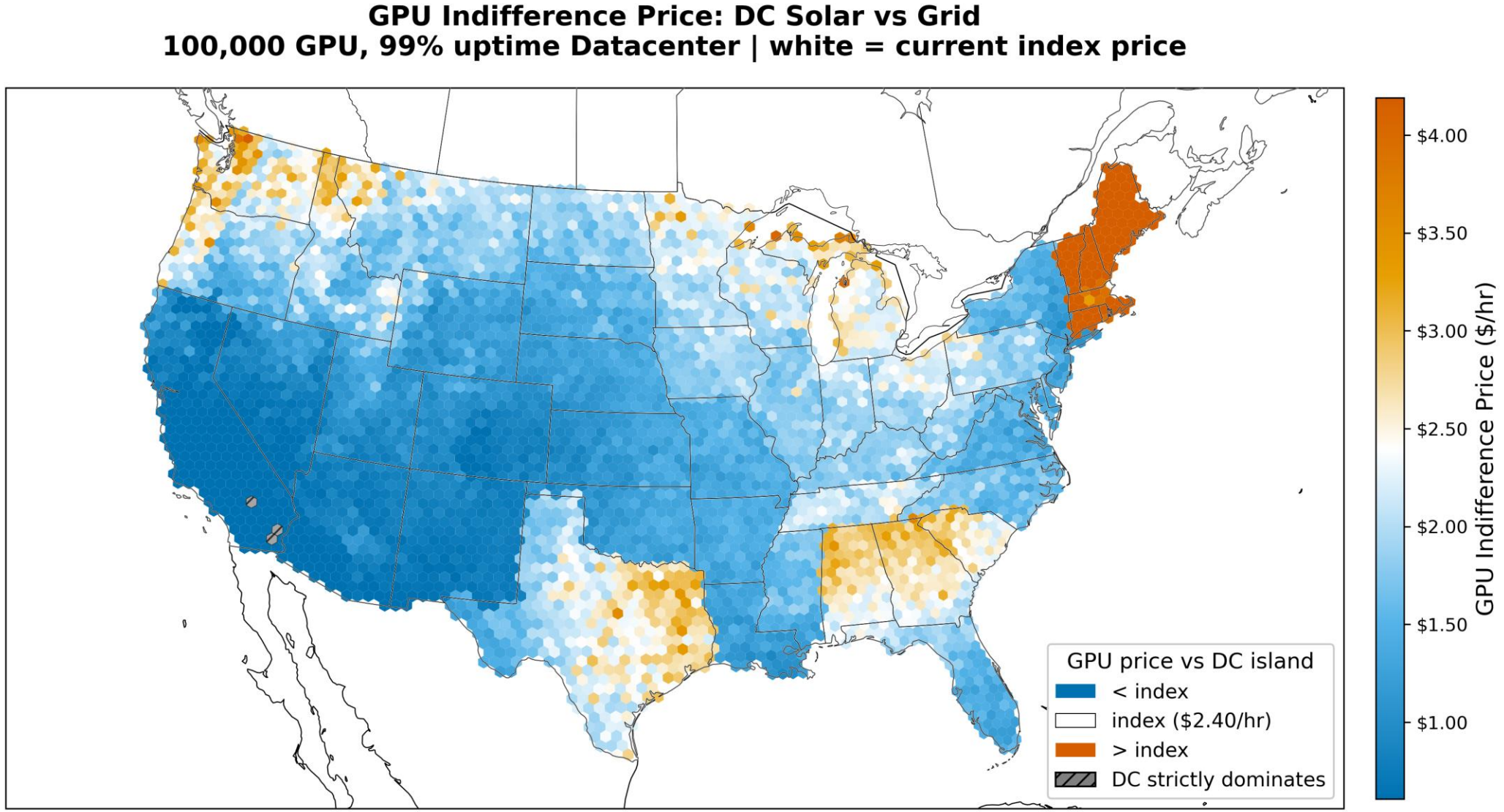


Note this comparison reflects like-for-like PV+BESS microgrid TCO. When comparing direct current ±400V data halls to AC equivalents each supplied with grid power, the all DC system has 7.9% lower total energy costs on average.

Results: deployment speed dominates direct energy cost considering the time value of energized chips



This finding is durable to significant reduction in GPU value across most locations



Tool Suite – Key Insights

- All-DC microgrids outperform conventional AC systems by 18% on average in total cost of ownership
- Storage systems sufficient to supply 99%+ uptime exhibit modest modeled capacity fade (~9% at replacement year) under empirically calibrated duty cycles
- When considering the time value of energized chips, All-DC microgrids outperform grid interconnection and generally outperform utility-scale natural gas

New Mexico Site Analysis
Location: 35.75°N, 104.72°W |IT **Design Load:** 30 MW
Uptime: 99.0% | **Natural Gas Price**: $4.58/MMBtu

System Type	Construction / Interconnect Time (Years)	LCOE (Base)	LCOE (w/ Chip Opp. Cost)
AC Solar + Storage (254 MW PV / 193 MW B)	2	$0.259/kWh	$0.690/kWh
DC Solar + Storage (241 MW PV / 168 MW B)	2	$0.212/kWh	$0.642/kWh
Natural Gas (70 MW)	4	$0.181/kWh	$1.13/kWh
Grid	4	$0.054/kWh	$1.01/kWh

Follow-up work: Varied facility operations, detailed reliability modelling, and additional generation technologies

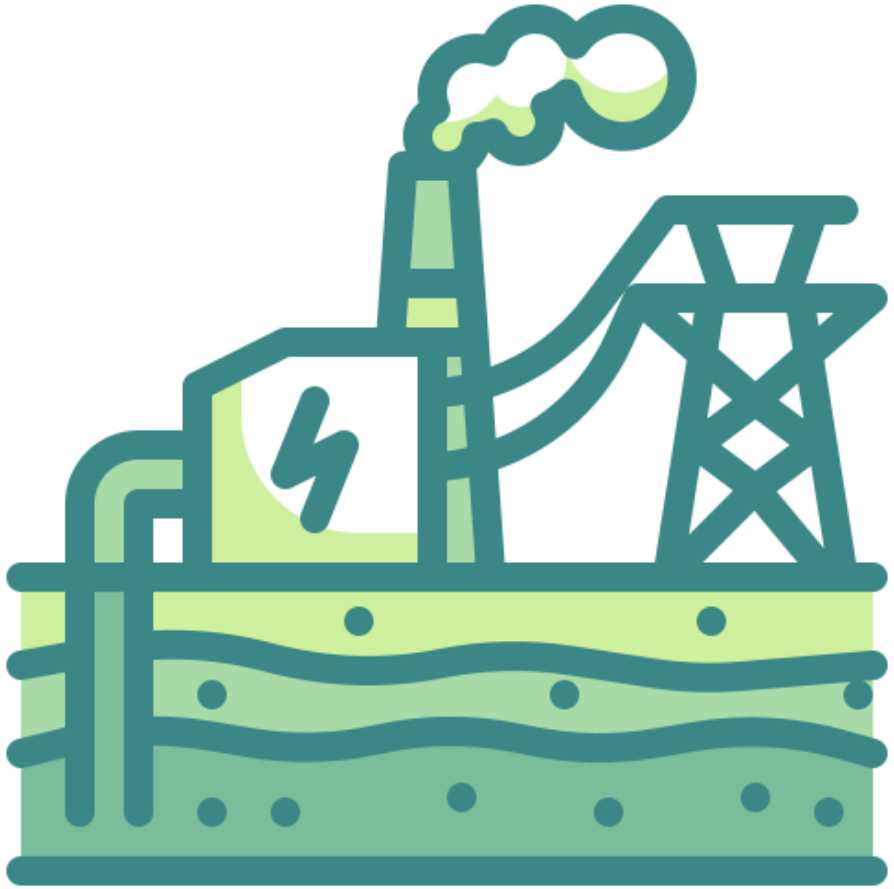


Image source: https://www.flaticon.com/free-icon/geothermal-energy_2317185

Modelling suite can be extended to various research questions:

- Demand response feasibility and efficacy
- Component-level reliability comparison between topologies
- Additional generation technologies: Reciprocating engines, fuel cells, wind, advanced geothermal
- Varied facility load profiles and dispatching strategies (e.g. inference vs training vs mixed use facility, different inference provisioning schedules)

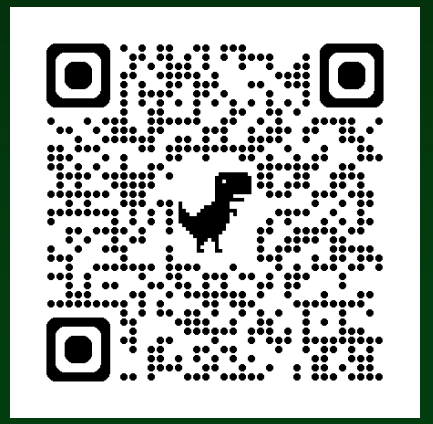
Calls-to-Action

How do we bring AI compute online given interconnection timelines and energy supply chains?

All-DC data center microgrids are fast to build and save 18% energy costs on average compared to AC equivalents, with DC data halls accounting for roughly half those savings

Contribute to OCP ongoing work to develop DC power distribution standards

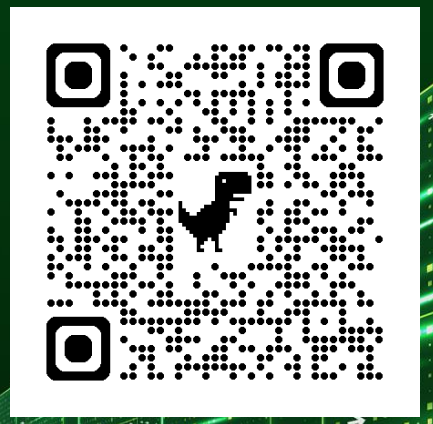
OCP Power distribution



OCP 400V DC sidecar training



Modelling Github Repository



Article preprint



Acknowledgements and References

Thanks to my coauthors Daniel L. Gerber, Erica R. H. Fuchs, Alex Hubbard, M. Granger Morgan, Arman Shehabi, Sarah J. Smith, and Constantine Samaras

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